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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

29 October 2026

Hiring Team
Aveni

Application: Senior AI Software Engineer (Remote, UK)

Dear Aveni hiring team,

I am applying for the Senior AI Software Engineer role. For three years at HDFC Asset Management I built and shipped LLM applications for regulated financial-services teams, owning them from the ReactJS frontend and FastAPI backend through RAG, tool-calling and AWS deployment. Your brief, engineering the systems and interfaces around models rather than training them, is the work I have been doing and want to keep doing.

At HDFC I designed and built the GenAI applications end to end: a ReactJS frontend served by Nginx, Python FastAPI microservices in Docker on AWS, and RAG with LlamaIndex, Azure OpenAI and a separate pgvector database per department so that access stayed siloed. Agents called Python functions as tools for SQL-backed fund comparisons, and the same logic ran as a WhatsApp bot used by roughly 50 to 100 Sales staff daily. When the model did not always extract the right parameters and tool calls failed, I fixed it by adding example inputs to the tool definitions. I released to real users and iterated on what broke, versioned prompts in Git, guardrailed behaviour through system prompts, had Compliance review each application, and allocated models by cost: self-hosted Llama-3 for the largest department and Azure OpenAI where output quality mattered.

That work was done in a regulated, high-trust setting: LDAP-based role access alongside department-based document access, traffic restricted to company private IPs, and a one-click Jenkins and GitLab CI/CD pipeline for dev and prod. I worked with the security team until the applications passed VAPT. A scheduled pipeline on SageMaker (EventBridge and Lambda) used Azure OpenAI to parse over 300 daily emails and saved 60 analyst hours a month.

I want to be direct about where I differ from your requirements. My production backend work is Python and FastAPI; I used Node.js and ExpressJS only during development at HDFC, and TypeScript on the CellLineSelector frontend, so I have not run a production Node.js and TypeScript backend or used NestJS. My AWS experience is EC2, ECR, S3, Lambda, EventBridge and SageMaker; I have not worked with ECS/Fargate, API Gateway, SQS/SNS or DynamoDB, so I have not built event-driven systems on messaging services. My agents were hand-built tool calling without LangGraph, and I have not used Langfuse; evaluation at HDFC was user feedback and Compliance review with prompts versioned in Git, not an eval framework or tracing. I am about three years in, with de facto lead experience over up to four interns rather than extensive seniority.

For my MSc dissertation with AstraZeneca I co-built CellLineSelector, where I designed the RAG layer and worked on the scoring algorithm and data preprocessing, built as FastAPI microservices with a React and TypeScript frontend. In my own Wingman project I treat LLM output as untrusted: responses are validated with Pydantic, malformed JSON is repaired, results are cached by input hash, and rate limits shaped the design, with results streamed to the UI. I work comfortably with product, compliance and engineering stakeholders to turn requirements into working tools.

I am happy to discuss working arrangements for a remote UK role. I hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane